

No. of Functional Features Included in the Solution

Date	15 July 2026
Team ID	TM-RISINGWATERS-04
Project Name	Rising Waters: AI Powered Flood Prediction System
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Interactive Map & Visualization	Renders raster inundation predictions on Leaflet.js with multi-layer overlays.	Frontend UI	Done	1.0
2	Deep Learning Inference Pipeline	Serves pre-trained UNet models using PyTorch to analyze elevation and predict flood maps.	ML Engine	Done	1.5
3	IoT Sensor Stream Intake	Processes live water telemetry streams from water-level sensors via WebSockets.	Backend Engine	Done	1.0
4	Automated SMS Broadcasts	Dispatches automated alerts to registered mobile phone numbers using Twilio APIs.	Notification Service	Done	0.5
5	Sentinel-2 Live Raster Pull	Trigger auto-ingestion of high-resolution cloud-free raw multispectral GeoTiffs.	Data Ingest	In Progress	0.5
6	Secure User Dashboard	Interactive admin console with multi-factor authentication for emergency dispatch roles.	Auth / Admin Panel	Done	0.5
7	Offline Support Routing	Computes local offline shortest path routing strategies if internet access drops during floods.	Geospatial Router	Pending	0.0
8	Telemetry History Export	Generates printable CSV / PDF reports containing sensor histories and predictions.	Reporting Hub	Done	0.0

Feature Summary:

Metric	Count / Value
Total Features Planned	8 Features
Total Features Implemented	6 Implemented (with 1 In Progress)
Core / Must-Have Features	5 Core Features (Map, ML Pipeline, IoT Intake, SMS Broadcast, Sentinel Ingest)
Additional / Nice-to-Have Features	3 Features (Dashboard, Offline Routing, Telemetry Export)
Features Tested & Verified	6 Features verified under local stress simulation suites

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Interactive Leaflet Map, Telemetry History Visualizer, Alert Settings Hub
2	Backend / Logic	2	UNet ML inference server, automated prediction mapping schedules
3	Database / Storage	1	Time-series logging for sensor metrics, PostgreSQL geospatial storage
4	API / Integration	2	Twilio SMS Broadcast platform, Sentinel-2 AWS S3 imagery access
5	Security / Authentication	1	Role-based JWT sessions, JWT access validation on API endpoints